NETFLOW

YERLİ ÜRETİM

التحكم في المستوى · ROD AND ROPE TYPE

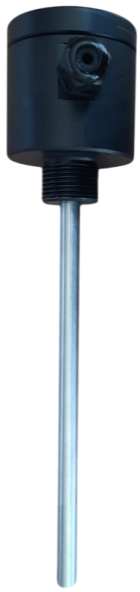
# ROD-TYPE DIGITAL
# المستشعر السعوي

High-performance level sensor that detects the presence of material in silos, bunkers, hoppers and pipes

This product detects whether product / material is present at the level where it is installed in silos, bunkers, hoppers, pipes, etc. Main target materials: grain, feed, powdered food, plastics, sand, liquids, etc. The output set point is adjusted with the trimpot, or with the display on display models; all parameters are configured using the 2-digit digital display and the MODE / ▲ / ▼ buttons on the board.

| | |
|---|---|
| **12 ... 30 VDC**<br>SUPPLY · MAX. 40 mA | **300 mA PNP / Relay**<br>OUTPUT · N.O. / N.C. SELECTABLE |
| **–40 ... +80 °C**<br>OPERATING TEMPERATURE · 70% RH | **INSERTION LENGTH**<br>STANDARD 200 / 300 / 400 mm (extendable on request) |

- Detection of liquids, powders and flakes
- Adjustable hysteresis and response time
- SS304 stainless steel rod, insulated from housing
- 4-step fine sensitivity adjustment
- Offset calibration for product residue
- Flange screw mounting

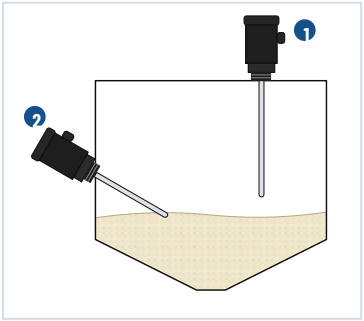


HOUSING VIEW

## المواصفات الفنية

| | |
|---|---|
| مبدأ القياس | Capacitive |
| Application | Liquids, powders, flakes / products |
| Supply | 12 ... 30 VDC |
| Current consumption | Max. 40 mA |
| الإخراج | P-type transistor (PNP) + output, max. 300 mA · N.O. / N.C. |
| Electrical connection | Cable gland entry, internal 16 – 28 AWG terminal / socket |

| | |
|---|---|
| Display | 2-digit digital display, output LEDs |
| Adjustment | Set trimpot · MODE / ▲ / ▼ buttons |
| Operating conditions | –40 °C ... +80 °C / max. 70% RH |
| Housing | Delrin |
| Rod | SS304 stainless steel rod, insulated from housing |
| Mounting | Flange, screw-fixed |

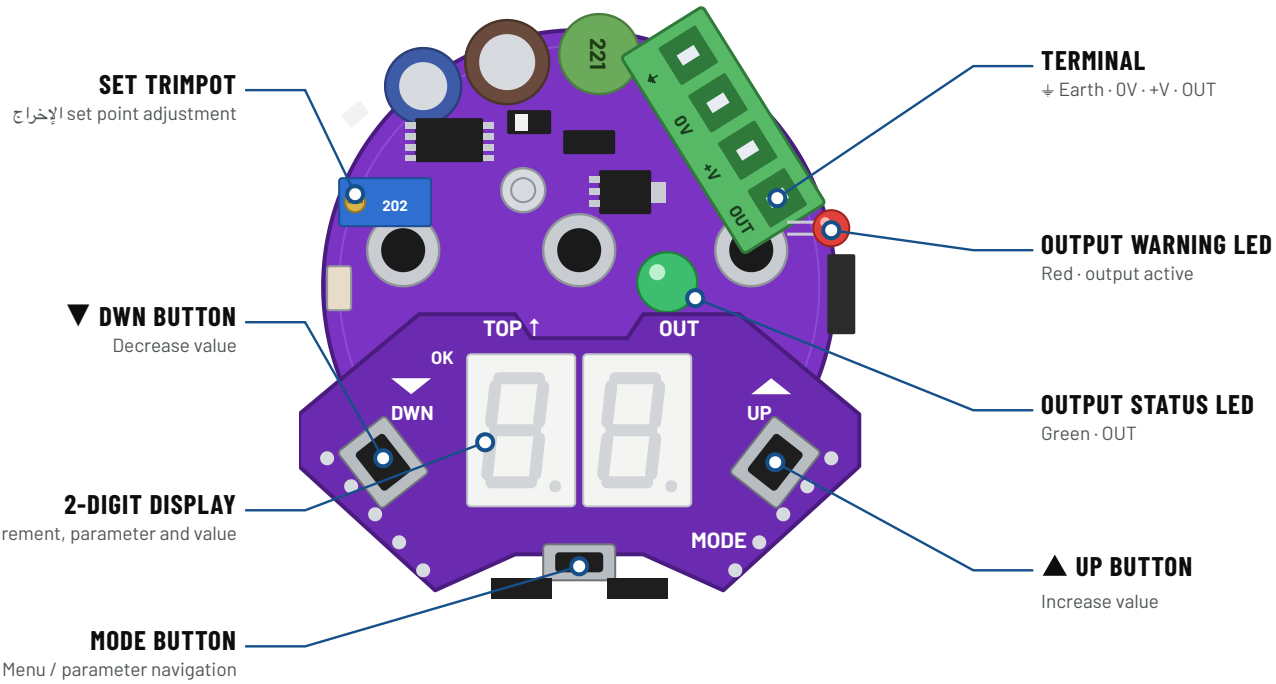



## التركيب AND APPLICATION

1. **Vertical mounting.** From the silo roof; the switching point is set by the rod length. Suitable for powder and flake applications.
2. **Downward-angled mounting.** In liquid applications, avoiding contamination at the housing joint improves measurement reliability.

**CAUTION** For liquid measurement, the power supply earth / PE must be isolated from the system chassis.

## INTERNAL VIEW DISPLAY MODEL · ELECTRONIC BOARD




### TERMINAL CONNECTION

| | |
|---|---|
| ⏚ | Earth (GND) |
| **0V** | Supply – (12 ... 30 VDC) |
| **+V** | Supply + (12 ... 30 VDC) |
| **OUT** | PNP + output, max. 300 mA |

### SETTINGS MENU

**01** Press and hold MODE for 4 – 5 s to enter the setting modes.

**02** The parameter number and its value are displayed in turn.

**03** Change the value with the ▲ / ▼ buttons.

**04** Press MODE repeatedly to move to the next parameter.

## PARAMETERS

| NO | PARAMETER | RANGE | DESCRIPTION |
|---|---|---|---|
| **P1** | **Set point** | 5 – 98 | In vertical applications, defines the output set point according to the position of the product in contact with the probe rod surface. |
| **P2** | **Threshold / hysteresis** | 1 – 20 | If required, a +/– threshold / hysteresis value is applied to the set point. |
| **P3** | **Fine sensitivity** | 1 – 4 | Set according to the specific density of the product. Min. sensitivity: liquids and similar · Max. sensitivity: low-density products such as flaked feed. |
| **P4** | **Response time filter** | 1 – 8 | Detection / output delay. 1: no delay · 2: ~1 s · 3: ~2 s · 4: ~3 s · 5: ~4 s · 6: ~5 s · 7: ~6 s · 8: ~7 s |
| **P5** | **الإخراج type** | N.O. / N.C. Relay / Transistor | Normally open or normally closed is selected. The output is a P-type transistor + output, max. 300 mA. |
| **P6** | **Offset calibration** | 21 / 10 | Sets the display to "0" when there is an obstacle or product residue build-up. |

**c.0.**

### CALIBRATION · P6 = 21

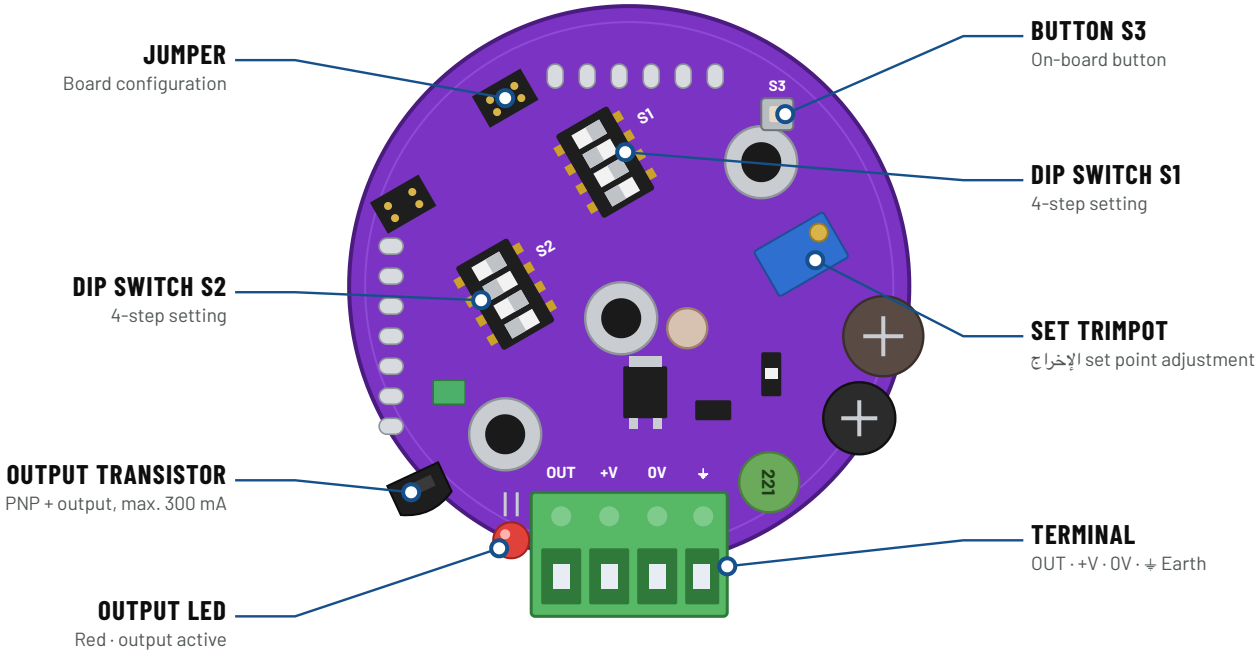
Exit with MODE or wait 10 – 15 s for automatic calibration. On success the display shows "c.o." (calibration OK).

**c.r.**

### FACTORY RESET · P6 = 10

Use the same procedure to cancel the calibration. On success the display shows "c.r." (calibration reset).

## INTERNAL VIEW TRANSISTOR MODEL · ELECTRONIC BOARD



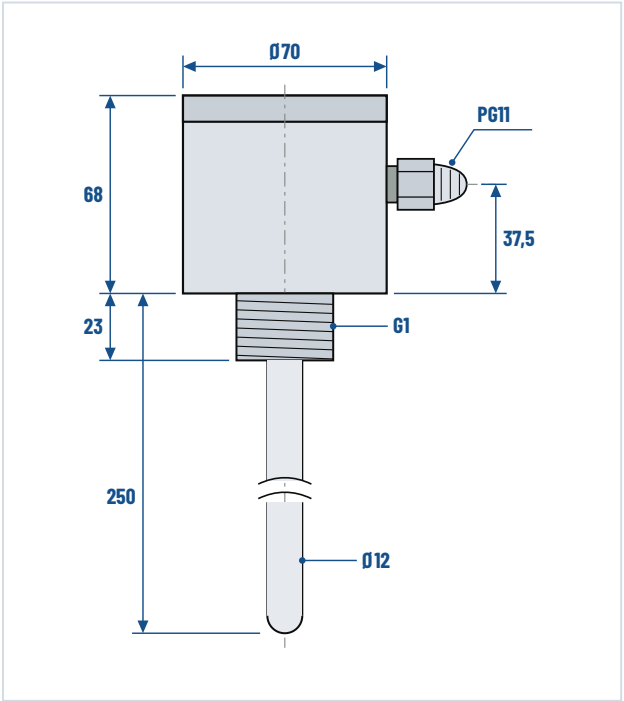

### TERMINAL CONNECTION

| | |
|---|---|
| **OUT** | PNP + output, max. 300 mA |
| **+V** | Supply + (12 ... 30 VDC) |
| **0V** | Supply – (12 ... 30 VDC) |
| ⏚ | Earth (GND) |

### ADJUSTMENT ELEMENTS

| | |
|---|---|
| **Set trimpot** | الإخراج set point adjustment |
| **S1 · S2** | 4-way DIP switch |
| **S3** | On-board button |
| **LED** | Red · output status |

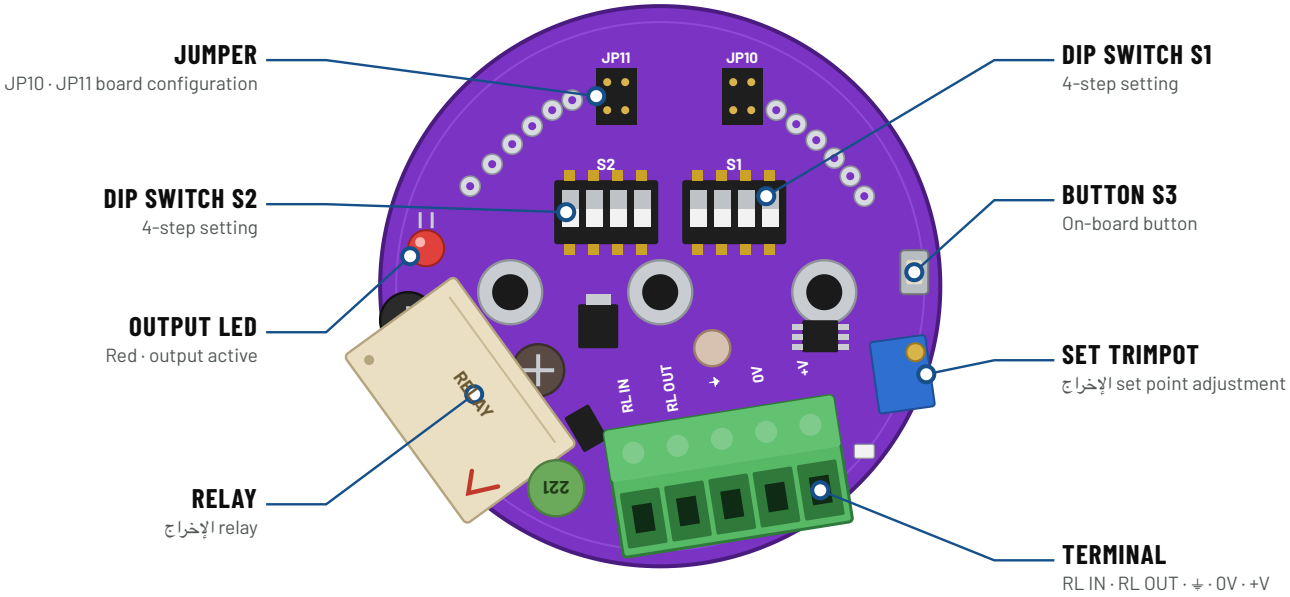
## الأبعاد TRANSISTOR MODEL · الأبعاد mm




| | |
|---|---|
| Housing diameter | Ø70 mm |
| Housing height | 68 mm |
| Cable entry | PG11 · 37.5 mm from housing bottom |
| وصلة العملية | G1 · 23 mm |
| Rod | Ø12 mm · 250 mm in drawing |

Rod length varies by order; all dimensions in mm.

## INTERNAL VIEW RELAY MODEL · ELECTRONIC BOARD



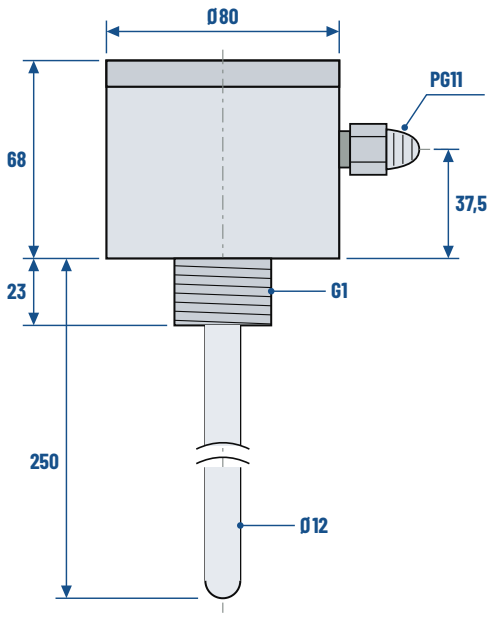

### TERMINAL CONNECTION

| | |
|---|---|
| **RL IN** | Relay contact terminal |
| **RL OUT** | Relay contact terminal |
| **⏚** | Earth (GND) |
| **0V** | Supply – (12 ... 30 VDC) |
| **+V** | Supply + (12 ... 30 VDC) |

### ADJUSTMENT ELEMENTS

| | |
|---|---|
| **Trimpot** | الإخراج set point adjustment |
| **S1 · S2** | 4-way DIP switch |
| **S3** | On-board button |
| **LED** | Red · output status |
| **Relay** | Contact output (RL IN – RL OUT) |

## الأبعاد RELAY MODEL · الأبعاد mm



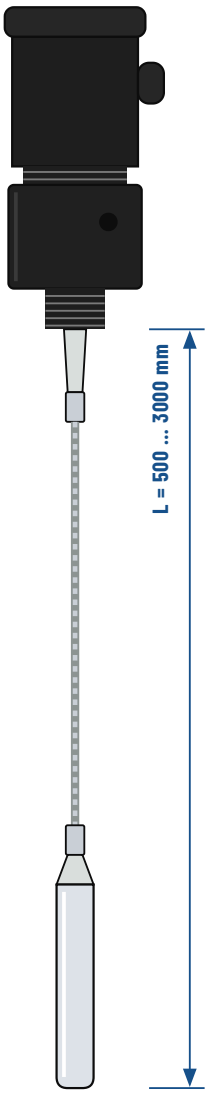

| | |
|---|---|
| Housing diameter | Ø80 mm |
| Housing height | 68 mm |
| Cable entry | PG11 · 37.5 mm from housing bottom |
| وصلة العملية | G1 · 23 mm |
| Rod | Ø12 mm · 250 mm in drawing |

Rod length varies by order; all dimensions in mm.

SUB-MODEL

# ROPE-TYPE المستشعر السعوي

Long-range level detection in tall silos and bunkers



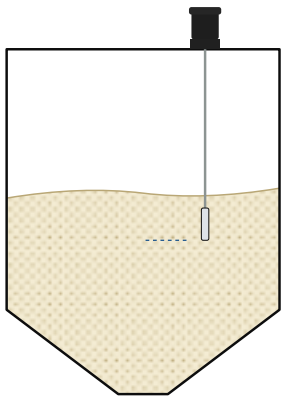

HOUSING VIEW

The rope type uses the same electronic head as the rod-type capacitive sensor; instead of a fixed rod, the sensing element is an end weight attached to a steel rope. The rope hangs vertically from the sensor housing into the silo, and the sensing point is lowered from the silo roof to the required depth. When the product reaches the end weight, the resulting change in capacitance is detected and the output is activated.

| | |
|---|---|
| **500 ... 3000 mm**<br>ROPE LENGTH · MIN. / MAX. | **Galvanized steel**<br>ROPE AND END WEIGHT |
| **SS316L / SS304**<br>OPTIONAL STAINLESS STEEL | **12 ... 30 VDC**<br>SUPPLY · SAME AS ROD MODEL |

## المواصفات الفنية

| | |
|---|---|
| مبدأ القياس | Capacitive |
| Sensing element | Steel rope + end weight |
| Rope length | Min. 500 mm · max. 3000 mm |
| Rope material | Galvanized steel · optional SS316L / SS304 |
| End weight material | Galvanized steel · optional SS316L / SS304 |
| Electronics | Display / transistor / relay model |
| Supply · output | Same as rod model (see page 01) |



## التركيب AND APPLICATION

1. **Vertical mounting.** The sensor is mounted on the silo roof; the rope must hang freely and vertically.
2. **Sensing point.** The rope length is selected between 500 – 3000 mm according to the depth at which the level is to be detected.
3. **Position.** The rope and end weight must not be below the filling stream or near areas where product can build up.

**CAUTION** Water ingress into the sensor housing must be prevented. See pages 02 – 04 for electrical connection and parameter settings.